Quantum Mechanics

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Chapter 1

Formalism

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Chapter 2

Simple Systems

2.1 Harmonic Oscillator

The Hamiltonian for one-dimensional harmonic oscillator is

$$H = \frac{1}{2m}p^2 + \frac{m\omega^2}{2}q^2. \tag{2.1.1}$$

We define

$$a = \frac{1}{\sqrt{2\hbar}}\Big(\sqrt{m\omega}q + i\frac{p}{\sqrt{m\omega}}\Big) \text{ and } a^\dagger = \frac{1}{\sqrt{2\hbar}}\Big(\sqrt{m\omega}q - i\frac{p}{\sqrt{m\omega}}\Big). \tag{2.1.2}$$

We can get

$$H = \hbar\omega\Big(a^\dagger a + \frac{1}{2}\Big) \tag{2.1.3}$$

and

$$[a, a^\dagger] = 1. \tag{2.1.4}$$

There is a long discussion to get the spectrum of this Hamiltonian...

The result is

$$E_n = \hbar\omega\Big(n + \frac{1}{2}\Big); \quad n = 0, 1, 2, \dots\ . \tag{2.1.5}$$

We define

$$x = \sqrt{\frac{m\omega}{\hbar}}q. \tag{2.1.6}$$

Then

$$\frac{d}{dx} = \sqrt{\frac{\hbar}{m\omega}}\frac{d}{dq} = i\frac{p}{\sqrt{m\hbar\omega}}. \tag{2.1.7}$$

The creation and annihilation operators in the x-representation are

$$a = \frac{1}{\sqrt{2}}\Big(x + \frac{d}{dx}\Big); \quad a^\dagger = \frac{1}{\sqrt{2}}\Big(x - \frac{d}{dx}\Big). \tag{2.1.8}$$

The lowest eigenstate satisfies $a\,|\varphi_0\rangle = 0$, meaning

$$\Big(x + \frac{d}{dx}\Big)\varphi_0(x) = 0 \tag{2.1.9}$$

the general solution is given by

$$\varphi_0(x) = c_0 \exp\left(-\frac{x^2}{2}\right), \tag{2.1.10}$$

c_0 is determined by the normalization condition

$$1 \stackrel{!}{=} \int_{-\infty}^{+\infty} dq|\varphi_0(q)|^2 = \sqrt{\frac{\hbar}{m\omega}}|c_0|^2 \int_{-\infty}^{+\infty} dx e^{-x^2} = \sqrt{\frac{\pi\hbar}{m\omega}}|c_0|^2. \tag{2.1.11}$$

We choose $c_0 = \left(\frac{m\omega}{\hbar\pi}\right)^{1/4}$. The higher eigenwavefunctions can be got from

$$\begin{aligned}\varphi_n(x) = \langle x|n\rangle = \frac{1}{\sqrt{n!}}\langle x|(a^\dagger)^n|0\rangle =& \frac{1}{\sqrt{2^n n!}}\Big(x - \frac{d}{dx}\Big)^n \varphi_0(x) \\ =& \frac{1}{\sqrt{2^n n!}}\Big(\frac{m\omega}{\hbar\pi}\Big)^{1/4}\Big(x - \frac{d}{dx}\Big)^n \exp\left(-\frac{x^2}{2}\right).\end{aligned} \tag{2.1.12}$$

The equation can be converted to the definition of the Hermite polynomials $\mathrm{H}_n(x)$. The relation is given by

$$\varphi_n(x) = \frac{1}{\sqrt{2^n n!}}\Big(\frac{m\omega}{\hbar\pi}\Big)^{1/4} \exp\left(-\frac{x^2}{2}\right)\mathrm{H}_n(x) \tag{2.1.13}$$

From (2.1.12), we get

$$\mathrm{H}_n(x) = \exp\left(\frac{x^2}{2}\right)\Big(x - \frac{d}{dx}\Big)^n \exp\left(-\frac{x^2}{2}\right), \tag{2.1.14}$$

which is equivalent to the definition of the Hermite polynomials:

$$\mathrm{H}_n(x) = (-1)^n e^{x^2}\frac{d^n}{dx^n}e^{-x^2}. \tag{2.1.15}$$

The equivalence can be seen by

$$
\begin{aligned}
&(-1)^n \exp\left(x^2\right) \frac{d^n}{dx^n} \exp\left(-x^2\right) \\
=&(-1)^n e^{x^2} \frac{d}{dx} e^{-x^2} e^{x^2} \frac{d}{dx} e^{-x^2} \cdots e^{x^2} \frac{d}{dx} e^{-x^2} \\
=&(-1)^n \left[e^{x^2} \frac{d}{dx} e^{-x^2} \right]^n \\
=&(-1)^n \left[e^{\frac{x^2}{2}} e^{\frac{x^2}{2}} \frac{d}{dx} e^{-\frac{x^2}{2}} e^{-\frac{x^2}{2}} \right]^n \\
=&(-1)^n \left[e^{\frac{x^2}{2}} \left(\frac{d}{dx} - x \right) e^{-\frac{x^2}{2}} \right]^n \\
=&e^{\frac{x^2}{2}} \left(x - \frac{d}{dx} \right) e^{-\frac{x^2}{2}} e^{\frac{x^2}{2}} \left(x - \frac{d}{dx} \right) e^{-\frac{x^2}{2}} \cdots e^{\frac{x^2}{2}} \left(x - \frac{d}{dx} \right) e^{-\frac{x^2}{2}} \\
=&e^{\frac{x^2}{2}} \left(x - \frac{d}{dx} \right)^n e^{-\frac{x^2}{2}}.
\end{aligned}
\tag{2.1.16}
$$

The only non-trivial step is the operator relation

$$
\frac{1}{f(x)} \frac{d}{dx} f(x) = \frac{d}{dx} + \frac{f'(x)}{f(x)}. \tag{2.1.17}
$$

Now we give the general solution of the one-dimensional harmonic oscillator, placed d from the original point,

$$
\varphi_n(q) = \frac{1}{\sqrt{2^n n!}} \left(\frac{m\omega}{\hbar\pi} \right)^{1/4} \exp\left(-\frac{(q+d)^2}{2l_\omega^2} \right) \mathrm{H}_n\left(\frac{q+d}{l_\omega} \right), \tag{2.1.18}
$$

where $l_\omega \equiv \sqrt{\frac{\hbar}{m\omega}}$.

2.2 Landau Level

The Lagrangian for a particle of charge $-e$ and mass m moving in a magnetic vector potential is

$$
L = \frac{1}{2} m \dot{\mathbf{x}}^2 - e \dot{\mathbf{x}} \cdot \mathbf{A}. \tag{2.2.1}
$$

The canonical momentum is

$$
\mathbf{p} = m\dot{\mathbf{x}} - e\mathbf{A}. \tag{2.2.2}
$$

The Hamiltonian is

$$
H = \dot{\mathbf{x}} \cdot \mathbf{p} - L = \frac{1}{2m} (\mathbf{p} + e\mathbf{A})^2. \tag{2.2.3}
$$

Define

$$\pi \equiv \mathbf{p} + e\mathbf{A} = m\dot{\mathbf{x}}. \tag{2.2.4}$$

Poisson brackets are

$$\{x_i, p_j\} = \delta_{ij} \text{ and } \{x_i, x_j\} = \{p_i, p_j\} = 0. \tag{2.2.5}$$

We can get

$$\{\pi_i, \pi_j\} = \sum_{k=1}^{3} \frac{\partial \pi_i}{\partial x_k}\frac{\partial \pi_j}{\partial p_k} - \frac{\partial \pi_i}{\partial p_k}\frac{\partial \pi_j}{\partial x_k} = -e\epsilon_{ijk}B_k. \tag{2.2.6}$$

The quantization of this system is given by

$$[x_i, p_j] = i\hbar\delta_{ij} \text{ and } [x_i, x_j] = [p_i, p_j] = 0. \tag{2.2.7}$$

The particle is constrained in the x-y plane and the magnetic field is set to $\nabla \times \mathbf{A} = B\hat{\mathbf{z}}$, the Hamiltonian is

$$H = \frac{1}{2m}(\pi_x^2 + \pi_y^2), \tag{2.2.8}$$

and we have[1]

$$[\pi_x, \pi_y] = -ie\hbar B. \tag{2.2.9}$$

If we define

$$a = \frac{1}{\sqrt{2e\hbar B}}(\pi_x - i\pi_y) \text{ and } a^\dagger = \frac{1}{\sqrt{2e\hbar B}}(\pi_x + i\pi_y), \tag{2.2.10}$$

we have

$$[a, a^\dagger] = 1 \tag{2.2.11}$$

Now the Hamiltonian is

$$H = \hbar\omega_B\left(a^\dagger a + \frac{1}{2}\right), \tag{2.2.12}$$

where $\omega_B \equiv eB/m$. This Hamiltonian has the same algebraic structure as that of harmonic oscillators. Thus, the spectrum is

$$E_n = \hbar\omega_B\left(n + \frac{1}{2}\right), \tag{2.2.13}$$

which is called the Landau level. Then we compute the eigenstates in two gauges respectively.

[1] We can see if we set $p = -\pi_y$ and $q = \pi_x/eB$, the Hamiltonian and commutator is just the same as that of the harmonic oscillator.

2.2.1 Landau Gauge

The Landau gauge is the choice

$$\mathbf{A} = xB\hat{\mathbf{y}}. \tag{2.2.14}$$

Now the Hamiltonian is

$$H = \frac{1}{2m}p_x^2 + \frac{1}{2}m\omega_B^2\Big(x + \frac{p_y}{eB}\Big)^2. \tag{2.2.15}$$

There exists translation invariance in y direction, meaning $[p_y, H] = 0$, which tells us p_y eigenvalues are good quantum number to characterize states. We can use the ansatz

$$\psi_k(x, y) = e^{iky} f_k(x). \tag{2.2.16}$$

Apply the Hamiltonian to this state, we get

$$H\psi_k(x, y) = \Big[\frac{1}{2m}p_x^2 + \frac{1}{2}m\omega_B^2\Big(x + \frac{\hbar k}{eB}\Big)^2\Big]\psi_k(x, y) \equiv e^{iky}H_k f_k(x), \tag{2.2.17}$$

where H_k can be interpreted as a harmonic oscillator with a displacement. By using the result (2.1.18), we get

$$\psi_k(x, y) \sim e^{iky}\mathrm{H}_n\Big(\frac{x + kl_B^2}{l_b}\Big)e^{-(x+kl_B^2)^2/2l_B^2} \tag{2.2.18}$$

where $l_B = \sqrt{\frac{\hbar}{m\omega_B}} = \sqrt{\frac{\hbar}{eB}}$.

2.2.2 Symmetric Gauge

The symmetric gauge is

$$\mathbf{A} = -\frac{yB}{2}\hat{\mathbf{x}} + \frac{xB}{2}\hat{\mathbf{y}}. \tag{2.2.19}$$

We have rotational symmetry about the origin in this choice of gauge, which means the angular momentum is a good quantum number. Actually, we will use the eigenvalues of $\frac{H}{\hbar\omega_B} + \frac{L}{\hbar} - \frac{1}{2} \equiv N_b$ to label the states.

$$\begin{aligned}
N_b =& \frac{1}{2m\hbar\omega_B}(\pi_x^2 + \pi_y^2) - \frac{1}{\hbar}(xp_y - yp_x) - \frac{1}{2} \\
=& \frac{1}{2\hbar eB}\Big[(p_x + eA_x)^2 + (p_y + eA_y)^2 - 2eB(xp_y - yp_x)\Big] - \frac{1}{2} \\
=& \frac{1}{2\hbar eB}\Big[(p_x + eA_x)^2 + (p_y + eA_y)^2 - 4(eA_y p_y + eA_x p_x)\Big] - \frac{1}{2} \\
=& \frac{1}{2\hbar eB}\Big[(p_x - eA_x)^2 + (p_y - eA_y)^2\Big] - \frac{1}{2} \equiv \frac{1}{2\hbar eB}(\tilde{\pi}_x^2 + \tilde{\pi}_y^2) - \frac{1}{2}
\end{aligned} \tag{2.2.20}$$

We can check that $[\tilde{\pi}_x, \tilde{\pi}_y] = ie\hbar B$. Define

$$b = \frac{1}{\sqrt{2e\hbar B}}(\tilde{\pi}_x + i\tilde{\pi}_y) \text{ and } b^\dagger = \frac{1}{\sqrt{2e\hbar B}}(\tilde{\pi}_x - i\tilde{\pi}_y), \tag{2.2.21}$$

we have

$$[b, b^\dagger] = 1. \tag{2.2.22}$$

And N_b can be written as

$$N_b = b^\dagger b. \tag{2.2.23}$$

The eigenvalues of N_b are $n_b \in \mathbb{N}$. The general states can be constructed from

$$|n, n_b\rangle = \frac{(a^\dagger)^n (b^\dagger)^{n_b}}{\sqrt{n! n_b!}} |0, 0\rangle , \tag{2.2.24}$$

where $a^\dagger$ and a are defined in the equation (2.2.10). The ground state $|0, 0\rangle$ is defined as $ab\,|0, 0\rangle = 0$.

We can see the energy eigenvalues only depend on n as $E_n = \hbar\omega_B(n + 1/2)$. And the angular momentum eigenvalues can be computed as

$$L\,|n, n_b\rangle = \hbar(n_b - n)\,|n, n_b\rangle . \tag{2.2.25}$$

Specifically, the angular momentum of the lowest landau level states is just n_b. The wavefunctions in the symmetric gauge can be computed in the z-representation, where $z = x + iy$. Now we use this new coordinate to express the operators:

$$\begin{aligned} a &= -i\sqrt{2}\Big(l_B\partial_{\bar{z}} + \frac{z}{4l_B}\Big), \quad a^\dagger = -i\sqrt{2}\Big(l_B\partial_z - \frac{\bar{z}}{4l_B}\Big); \\ b &= -i\sqrt{2}\Big(l_B\partial_z + \frac{\bar{z}}{4l_B}\Big), \quad b^\dagger = -i\sqrt{2}\Big(l_B\partial_{\bar{z}} - \frac{z}{4l_B}\Big). \end{aligned} \tag{2.2.26}$$

The lowest landau level wavefunction satisfies $a\psi_{LLL}(z, \bar{z}) = 0$, which gives

$$\psi_{LLL}(z, \bar{z}) = \mathcal{N} e^{-\frac{|z|^2}{4l_B^2}} f(z), \tag{2.2.27}$$

where $\mathcal{N}$ is some normalization constant and $f(z)$ is an arbitrary holomorphic function. The equation $b\psi_{LLL}(z, \bar{z}) = 0$ gives $f(z) = \text{const}$. Thus the ground state is

$$\psi_{0,0}(z, \bar{z}) = \frac{1}{\sqrt{2\pi l_B^2}} e^{-\frac{|z|^2}{4l_B^2}}. \tag{2.2.28}$$

The higher angular momentum states in the lowest landau level can be constructed

by applying $b^\dagger$ multiple times:

$$\psi_{0,n_b}(z,\bar{z}) = \frac{(b^\dagger)^{n_b}}{\sqrt{n_b!}}\psi_{0,0}(z,\bar{z}) = \frac{1}{\sqrt{2\pi l_B^2 n_b!}}\left(\frac{z}{\sqrt{2}l_B}\right)^{n_b} e^{-\frac{|z|^2}{4l_B^2}}. \tag{2.2.29}$$

We can check the angular momentum eigenvalue of this state is indeed

$$L = \hbar(z\partial_z - \bar{z}\partial_{\bar{z}})\psi_{0,n_b} = \hbar n_b \psi_{0,n_b}. \tag{2.2.30}$$

This result is important in the study of fractional quantum Hall effect.

2.A Computations

2.A.1 The Gaussian Integrals

The simplest Gaussian integral is

$$\int_{-\infty}^{+\infty} dx e^{-x^2} = \sqrt{\pi}, \quad a > 0. \tag{2.A.1}$$

The technique of computing more complicated Gaussian integrals from the (2.A.1) is very useful.

$$\begin{aligned}\int_0^{+\infty} dx x^{2n} e^{-\alpha x^2} =& (-1)^n \frac{d^n}{d\alpha^n}\int_0^{+\infty} dx e^{-\alpha x^2} \\ =& (-1)^n \frac{d^n}{d\alpha^n}\frac{1}{2}\sqrt{\frac{\pi}{\alpha}} \\ =& \sqrt{\frac{\pi}{\alpha}}\frac{(2n-1)!!}{2^{n+1}\alpha^n}.\end{aligned} \tag{2.A.2}$$

And if we compute the product of two simple Gaussian integrals (2.A.1), we can get

$$\begin{aligned}\frac{\pi}{a} = \left(\int_{-\infty}^{+\infty} dx e^{-\alpha x^2}\right)^2 =& \int_{-\infty}^{+\infty} dx \int_{-\infty}^{+\infty} dy e^{-a(x^2+y^2)} \\ =& \int_0^{2\pi} d\theta \int_0^{\infty} r e^{-ar^2}.\end{aligned} \tag{2.A.3}$$

We can get

$$\int_0^{+\infty} dr r e^{-ar^2} = \frac{1}{2a}. \tag{2.A.4}$$

Then we can compute

$$\begin{aligned}\int_0^{+\infty} dx x^{2n+1} e^{-\alpha x^2} =& (-1)^n \frac{d^n}{d\alpha^n} \int_0^{+\infty} dx x e^{-\alpha x^2} \\ =& (-1)^n \frac{d^n}{d\alpha^n} \frac{1}{2\alpha} \\ =& \frac{n!}{2\alpha^{n+1}}.\end{aligned} \tag{2.A.5}$$

Chapter 3

Angular Momentum

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Chapter 4

Central Potential

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Chapter 5

Approximation Methods

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Chapter 6

Scattering Theory